

A covariant, ghost-free modified-inertia completion of the horizon-tied acceleration scale $a_0 = cH_\Lambda/Z$, and a passivity obstruction to single-metric MOND lensing

Carl P. Zimmerman

Briar Creek Tech (independent researcher)

carl@briarcreektech.com · ORCID [0009-0008-3508-7982](https://orcid.org/0009-0008-3508-7982)

July 2026

Abstract

The de Sitter–Unruh modified-inertia (MI) program reproduces galaxy dynamics through a passive vacuum-response kernel $K(\Box_u/a_0^2)$ that ties the MOND acceleration scale to the cosmological horizon, $a_0 = cH_\Lambda/Z$ with $Z = \sqrt{32\pi/3}$. We present a single covariant action realizing this program — Einstein–Hilbert gravity, a passive unit-timelike frame with zero propagating degrees of freedom, and a matter sector dressed by a bounded, causal, positive-measure (Herglotz–Nevanlinna) kernel — and show that the coupled system is ghost-free and well posed, conditional on the passive-frame premise. The interpolating kernel $\nu = \sqrt{1 + 1/y}$ is Milgrom’s (1999) and the $a_0 \propto \sqrt{\rho_{\text{DE}}}$ scaling is Limbach–Psaltis–Özel’s (2008); the distinctive content is the cH_Λ/Z coefficient and the covariant passive-frame completion, with the value of a_0 , the constant Z , and the sign $s = -1$ postulated, not derived. Within this theory we prove an obstruction: conditional on kernel passivity, a horizon-tied a_0 (D) and single-metric propagation with $c_\gamma = c_{\text{GW}}$ (S) forbid single-metric MOND weak lensing (L): $D \wedge S \Rightarrow \neg L$, with $\{D, S, L\}$ the unique minimal unsatisfiable subset of $\{D, S, G, L\}$ and ghost-freedom G entering no step. The mechanism is a passivity/amplification dichotomy: a passive kernel ($\sup K = 1$) can only suppress a source to ρ/ν , whereas the MOND-lensing phantom demands enhancement to $\nu\rho$. Completions reproducing lensing exist, but only as modified gravity with a_0 demoted to a free coupling. All results are reported on both cosmological footings (9.36×10^{-11} and $1.13 \times 10^{-10} \text{ m s}^{-2}$); the obstruction is footing-independent, and every load-bearing claim traces to a committed, runnable script.

PhySH: modified gravity; alternatives to general relativity; gravitational lensing; dark matter.

1 Introduction

1.1 The framework, and the credit

Modified inertia (MI) treats MOND phenomenology not as a modification of the gravitational field sourced by matter, but as a modification of the inertial response of matter to a field. Any honest statement of this program begins with the credit. Milgrom’s original MOND is Ref. [1]. In 1999 Milgrom observed [2] that a non-inertial observer in a de Sitter universe sees an Unruh temperature $T \propto \sqrt{a^2 + c^2 H_\Lambda^2}$, and that the excess of that temperature over the inertial-observer floor defines an acceleration function with exactly the MOND limits; his Eq. (9) contains the interpolating modified-inertia kernel used here, machine-verified identical. The $a_0 \propto \sqrt{\rho_{\text{DE}}}$ scale coincidence

likewise predates this work: it is Limbach, Psaltis & Özel [3]. The framework’s distinctive content is precisely (i) the coefficient — $a_0 = cH_\Lambda/Z$ with $Z = \sqrt{32\pi/3} = 5.789$, a factor $2Z \approx 11.58$ below Milgrom’s $\hat{a}_0 = 2cH_\Lambda$ — and (ii) the covariant modified-inertia completion built on a passive frame, which Milgrom’s 1999 note explicitly left open (he declined to endorse the kernel for circular orbits absent a modified-inertia formulation).

In the de Sitter–Unruh realization, a passive cosmic rest frame u^μ — a unit-timelike vector carrying **zero** propagating degrees of freedom, fixed by a Lagrange constraint $(u \cdot u + 1) = 0$ — mediates a vacuum response whose kernel acts on the frame d’Alembertian $\square_u$. The worldline law that emerges is

$$g_{\text{obs}} = \nu(y) g_{\text{bar}}, \quad \nu(y) = \sqrt{1 + \frac{1}{y}}, \quad y \equiv \frac{g_{\text{bar}}}{a_0}, \quad (1)$$

reducing to the deep-MOND law $g_{\text{obs}} \rightarrow \sqrt{a_0 g_{\text{bar}}}$ as $y \rightarrow 0$ and to Newton as $y \rightarrow \infty$. This fits the SPARC rotation-curve sample [13] at 0.108 dex scatter ($\Upsilon = 0.70$), competitive with regularized MOND (reproducible via the companion tool `a0kit` [27]).

Throughout, *footing* denotes the choice of cosmological input to $a_0 = cH_\Lambda/Z$: the **canonical** pure- Λ footing ($\rho_{\text{DE}}, cH_\Lambda/Z$) gives $a_0 = 9.36 \times 10^{-11} \text{ m s}^{-2}$, and the **alternate** footing ($\rho_{\text{total}}, cH_0/Z$) gives $1.13 \times 10^{-10} \text{ m s}^{-2}$. The two differ by 21% and are not separated by any current measurement; both are carried on every a_0 -valued number below. Signature $(-, +, +, +)$.

1.2 Honest scope of the claim

We are precise about what is and is not derived. What the framework establishes is the **structural tie of the scale of a_0** to the passive kernel and the de Sitter horizon. The **numerical value of a_0** , the constant $Z = \sqrt{32\pi/3}$, and the **sign $s = -1$** of the coupling are **postulates** — the program is a one-parameter effective-field-theory *reframing*, not a first-principles computation ($\kappa = 1/2$ is provably unforceable from ghost-freedom, unitarity, and holography). The ν -kernel law itself is Milgrom’s [2], and the $\sqrt{\rho_{\text{DE}}}$ coincidence is Limbach–Psaltis–Özel’s [3]. We credit both explicitly and do not re-assert any earlier, publicly retracted claims of a derived value of a_0 , a Standard-Model bridge, or a final all-physics theory. The defensible position is the a_0 reframing, and nothing more. A condensed derived-versus-postulated ledger is given in Appendix B.

1.3 Related work by the author

This manuscript consolidates four of the author’s Zenodo deposits, released as the timestamped preprint record for reproducibility: the MI field-theory results [14] (with erratum [15]), the covariant-completion derivation [16], the passivity/lensing obstruction [17], and the adversarially reviewed honest-scope no-go paper [18]. This is the standard preprint disclosure, not duplicate publication; no deposit is the primary source of more than one journal submission. Two companion submissions treat the phenomenology and are *not* consolidated here: the cosmological $a_0(z)$ pre-registered test (companion, JCAP submission; deposits [22, 23, 24]) and the SPARC a_0 -line slope measurement with the Λ inversion (companion, MNRAS submission; deposit [25]). The present paper is confined to the field-theory construction and the obstruction theorem.

One statement in Ref. [18] is superseded by the present construction and must be flagged: its abstract asserts that no covariant modified-inertia completion exists in three exhaustive cases. That no-go rested on the Einstein-aether strong-coupling wall, which assumes a *propagating* aether; it does not apply to the passive frame constructed here, which carries zero propagating modes (the correction is recorded in Ref. [19]). What is consolidated from Ref. [18] is its honest-scope ledger and its reframing algebra, which remain valid.

1.4 The lensing question, and the statement of results

Galaxy dynamics is a worldline phenomenon; weak lensing is not. Lensing probes the deflection of null geodesics, fixed by the gravitating stress tensor that the field equations actually see. The measured weak-lensing radial-acceleration relation (RAR) of Brouwer et al. [10] (KiDS-1000 isolated galaxies) coincides with the *dynamical* RAR: null geodesics feel an effective lensing mass enhanced to $\nu\rho$. As shown below, the assembled MI stress tensor sources the *suppressed* density ρ/ν ; single-metric pure MI is excluded relative to the lensing-RAR = dynamical-RAR equality at $\sim 27\sigma$ (conservative rail, footing-independent). The gap between what is sourced and what the equality requires is a factor ν^2 .

This paper contains two results, presented as one story. First (Secs. 2–4), the *positive* construction: a single covariant action realizing the MI program, with its degree-of-freedom count, kernel rigor, stress tensor, and ghost analysis, machine-verified. Second (Secs. 5–8), the *negative* structural result: conditional on kernel passivity, the completion question — can the ν^2 wedge be closed by adding structure while retaining everything that makes the program distinctive? — is answered no. The obstruction theorem is

Assume kernel passivity. Then $D \wedge S \Rightarrow \neg L$: no field theory extending the MI action keeps a_0 tied to the passive kernel (D), propagates photons and gravitons on one metric (S), and reproduces the single-metric MOND-lensing phantom $(\nu-1)\rho$ (L). Equivalently $\{D, S, L\}$ is the unique minimal unsatisfiable subset of $\{D, S, G, L\}$; ghost-freedom G enters no step.

The physical crux is a passivity/amplification dichotomy (Sec. 8): the property that derives a_0 's scale (passivity, $\sup K = 1$) is the very property that forbids the lensing enhancement ($\nu > 1$, outside the passive cone). A completion *does* exist — but only as modified *gravity*, at the cost of a_0 becoming a free coupling. We frame this as neither a triumph nor a funeral: it is an exact obstruction, conditional on passivity, that shows the a_0 reframing is load-bearing, and it redirects the search for an MI-versus-MG discriminant away from lensing.

2 The covariant action

2.1 The single functional

One metric $g_{\mu\nu}$, one passive unit-timelike frame u^μ (the cosmic/dS-Unruh rest frame), one Lagrange multiplier λ , matter ψ , one kernel K , one scale a_0 :

$$S[g, u, \lambda, \psi] = S_{EH}[g] + S_u[g, u, \lambda] + S_{\text{matter}}[g, u, \psi; K], \quad (2)$$

with

$$S_{EH} = \frac{c^4}{16\pi G} \int \sqrt{-g} R \quad (\text{host gravity unmodified: 2 graviton dof}), \quad (3)$$

$$S_u = - \int \sqrt{-g} \frac{\lambda}{2} (u^\mu u_\mu + 1) \quad (\text{passive frame, no kinetic term: 0 dof}), \quad (4)$$

$$S_{\text{matter}} = - \frac{1}{2} \int \sqrt{-g} \rho_m [s u^\mu K(\square_u/a_0^2) u_\mu] \quad (\text{modified inertia, in the matter kinetic sector}), \quad (5)$$

where

$$K(z) = \frac{\sqrt{1+4z}-1}{2\sqrt{z}}, \quad \square_u f = u^a \nabla_a (u^b \nabla_b f), \quad s = -1 \text{ (postulate)}. \quad (6)$$

The multiplier is fixed algebraically on shell, $\lambda = -\rho_m s K$: no new field, no new free function, no new propagating degree of freedom.

Kernel rigor (derived). K is a Herglotz–Nevanlinna function with a *unique* positive Borel measure: closed-form densities $\rho_A = (1 - \sqrt{1 - 4|t|})/(2\pi\sqrt{|t|})$ on $-1/4 < t < 0$ and $\rho_B = 1/(2\pi\sqrt{|t|})$ on $t < -1/4$, additive constant $a = 0.65411$, $\|K\| \leq 1$, causal-retarded [20, 21]. The sum rule

$$\int \frac{d\mu(t)}{|t|} = K(\infty) - K(0) = 1 \quad (\text{region-B share } 2/\pi \text{ exact; total } = 1 \text{ to } 10^{-8}) \quad (7)$$

with $K(0) = 0$ means a_0 enters *only* through the kernel argument and carries no counterterm at scale a_0 . Passivity — a positive spectral measure with unit total weight — is what floors the response amplitude at the horizon scale cH_Λ ; it is the source of the scale-tie, and it is the **single premise** of the obstruction result below (Sec. 9).

2.2 The worldline law: first-moment reduction

The bridge from the nonlocal operator to phenomenology is the first-moment identity [16]

$$u_\mu \square_u u^\mu = -|a|^2 \quad (8)$$

re-derived three independent ways (flat general worldline; general curved metric, symbolic, arbitrary g_{ab} and full connection; static Schwarzschild observer in closed form). The only inputs are unit norm $u \cdot u = -1$ and metric compatibility $\nabla g = 0$: no circularity, no geodesy, no spacetime symmetry. Hence $K(\square_u/a_0^2) \rightarrow K(X)$ with $X = |a|^2/a_0^2$ and $a^\mu = u^b \nabla_b u^\mu$ — the acceleration enters as a first-order *field* gradient, not a worldline $\ddot{x}$, which is how the classic modified-inertia Ostrogradsky trap is evaded.

On the rotation-curve shell the kernel argument collapses. With the on-shell condition $|a| = g_{\text{obs}} = \nu g_{\text{bar}}$ (so $X = y^2 + y$), one has $1 + 4X = (2y + 1)^2$, and the radical collapses:

$$K|_{\text{on-shell}} = \frac{(2y + 1) - 1}{2\sqrt{y^2 + y}} = \frac{1}{\nu(y)} \text{ exactly,} \quad \frac{2K'X}{K}|_{\text{on-shell}} = \frac{1}{2y + 1} \leq 1, \quad (9)$$

giving the ring-exact law Eq. (1) with ring residual $\sim 10^{-13}$, both footings. The on-shell dressing is the *suppression* $1/\nu \leq 1$, and the anisotropic slip factor $2K'X/K = 1/(2y + 1)$ is bounded and tension-signed. The static real-space dressing $K(X)$ is a bounded, increasing (Bernstein-type) function: $K(0^+) = 0$ at the deep-MOND end and

$$\boxed{\sup_{X \geq 0} K(X) = 1} \quad (\text{Newtonian saturation; } K \rightarrow 0 \text{ deep-MOND}), \quad (10)$$

saturated as $X \rightarrow \infty$. ($K(X)$ is not completely monotone — completely monotone functions decrease; here K increases and K' is completely monotone, the Bernstein/Stieltjes dual. The Herglotz–Nevanlinna, $\|K\| \leq 1$, and sum-rule properties attach to the frequency-domain response kernel $K(\square_u)$; the static $K(X)$ is the corresponding real-space dressing.) The single fact the obstruction uses downstream is the uniform bound (10): the passive kernel can only ever *suppress* a source to $\rho K \leq \rho$.

3 Degrees of freedom and ghost-freedom

The full coupled census (`wellposed.py`, 28/28 checks, exit 0, both footings):

property	verdict
propagating content	2 graviton + standard matter; frame = 0
Ostrogradsky ghost	none: local action first order in every dynamical field; nonlocal K ghost-free by the Herglotz single healthy pole
Hamiltonian bounded below	yes: $\mu = K \in (0, 1]$
hyperbolicity	block-diagonal principal symbol, real characteristics; well-posed Cauchy problem (retarded history data)
causality	retarded kernel; single global causal order
c_T	= 1 exact (the graviton rides g)

Three points deserve emphasis.

The frame stays passive under the matter coupling. The unit-norm second-class Dirac pair $\chi_1 = u \cdot u + 1$, $\chi_2 = u \cdot \pi$ has bracket $\{\chi_1, \chi_2\} = 2(u \cdot u)$, block determinant $4(u \cdot u)^2 \rightarrow 4$ on shell — a *kinematic* bracket, unchanged when the momentum carries an arbitrary matter contribution, with no tertiary constraint tower. The frame principal symbol $(u \cdot k)^2 \rightarrow k_0^2$ is a transport ODE (double root at $k_0 = 0$ independent of spatial k , zero group velocity): the $K(a^2)$ dressing structurally cannot promote u to a propagating aether. This machine-verified second-class Dirac closure is what keeps the frame at zero degrees of freedom even after the matter coupling.

The nonlocal sector was checked for real — with a precisely scoped residual. An earlier machine check of the nonlocal sector was found to be tautological and was replaced by a genuine verification (`mi_closure_pin/ostro_nonlocal_verify.py`, 13/13, no hard-coded booleans): the exact Herglotz spectral representation gives an auxiliary tower with Ostrogradsky Hessian $\partial^2 L / \partial \ddot{\chi}^2 = 0$, kinetic Hessian $2 d\mu(t) > 0$, and positive masses — healthy massive scalars, not a ghost — and the same machinery correctly *flags* a textbook $\ddot{q}^2$ ghost and a negative-measure kernel (live negative controls), so the check is discriminating, not decorative. This auxiliary-tower Herglotz ghost check is machine-verified; what remains *asserted, not verified*, is the fully coupled all-orders nonlocal Hamiltonian (and, for the disformal witness of Sec. 7, global $B < 1$ off spherical symmetry). Both residuals are carried in the ledger (Appendix B).

The honest hinge. The 0-dof + ghost-free verdict rests on the passive/hypersurface-orthogonal frame premise. A dynamical khronon T would put $a \sim \partial^2 T$ inside K and reopen an Ostrogradsky concern. The premise is the framework’s postulate, stated as such — it is the load-bearing hinge of the construction, not a proven theorem. Crucially, the obstruction theorem below does not depend on ghost-freedom (G enters no step), so these residuals do not affect the paper’s central result.

4 The stress tensor

4.1 Assembly, WEP, conservation

Matter couples minimally to the single metric g ; modified inertia is a universal inertial-scalar dressing $W = s u^\mu K(\Box_u/a_0^2) u_\mu$ of the matter kinetic term. The metric variation gives the closed-form matter stress tensor

$$\boxed{T_{\mu\nu} = \alpha u_\mu u_\nu + \beta g_{\mu\nu} + \gamma a_\mu a_\nu}, \quad \alpha = -\rho_m s K, \quad \beta = \frac{1}{2} \rho_m s K, \quad \gamma = \frac{\rho_m s K'(X)}{a_0^2}. \quad (11)$$

In the principal (UV) limit $K \rightarrow 1$, $K' \rightarrow 0$: the anisotropic term vanishes and $T_{\mu\nu}$ is an isotropic perfect fluid — no gravitational slip ($\Psi = \Phi$), the AeST no-slip property recovered from modified *inertia* with no dynamical aether. The only slip-capable stress is $\gamma a_\mu a_\nu$, nonzero only in the MOND/IR regime.

WEP is exact: $\eta = 0$, **derived.** W is built only from $(u, g, \partial u)$ and carries no matter-species label; every body's inertial coefficient is the same function $K(X)$, and the balance inverts species-independently (machine check: $\eta < 10^{-12}$ over $y \in [10^{-2}, 10^2]$; `matter_coupling_Tmunu.py`, 14/14, exit 0).

Conservation, verified two ways. $\nabla_\mu T^{\mu\nu} = 0$ on shell by diffeomorphism invariance: (i) the canonical Noether identity holds identically (symbolic, exact); (ii) instantiated with the actual MI acceleration kernel, the residual is 2.6×10^{-16} at 12 random spacetime points. The unit-norm constraint is dynamically preserved ($u \cdot a = 0$); the frame stays passive under the matter coupling.

4.2 The assembled tensor and the ν^2 wedge

Assembling matter and frame legs in the only Newton-consistent bookkeeping (the composite frame $u = J^\mu/|J|$, for which $u \cdot u = -1$ identically, so the frame-constraint leg $S_u \equiv 0$ and its stress $T^u \equiv 0$; the $1/\nu$ dressing therefore comes entirely from the *matter* leg ρK , not from a frame leg) yields

$$\hat{T}_{\mu\nu} = \rho K(X) u_\mu u_\nu - 2 \frac{\rho K'(X)}{a_0^2} a_\mu a_\nu. \quad (12)$$

On the RAR shell this sources $\rho_{\text{eff}} = \rho K = \rho/\nu$ (a *suppression*) plus an anisotropic radial tension $\Pi = -\rho_{\text{eff}}/(2y + 1)$ carrying the bounded, tension-signed slip factor — it *reduces* lensing rather than enhancing it. Linearizing Einstein on the single metric ($ds^2 = -(1 + 2\Phi)dt^2 + (1 - 2\Psi)d\vec{x}^2$),

$$\nabla^2 \Psi = 4\pi G \rho_{\text{eff}}, \quad \Phi' - \Psi' = 4\pi G r \Pi, \quad g_{\text{lens}} = \frac{1}{2}(\Phi' + \Psi'), \quad (13)$$

every source term is $O(K) \leq 1$. There is **no** $O(\nu)$ **structure anywhere** in the assembled tensor. The single metric therefore under-lenses:

$$F(y) \equiv \frac{g_{\text{lens}}}{\nu(y) g_{\text{bar}}} < \frac{1}{\nu(y)} < 1 \quad \text{everywhere.} \quad (14)$$

The **wedge** is the gap between what lensing needs and what MI sources. The observed lensing-RAR = dynamical-RAR equality [10] needs effective mass $\nu\rho$ (enhanced *up*), while MI supplies ρ/ν (suppressed *down*):

$$\underbrace{\nu\rho}_{\text{needed}} \quad \text{vs} \quad \underbrace{\rho/\nu}_{\text{sourced}} \quad \implies \quad \text{factor } \nu^2 = 1 + \frac{1}{y}. \quad (15)$$

The missing piece is the **phantom mass** $M_{\text{ph}} = (\nu - 1)M_{\text{bar}}$, exactly the density that QUMOND/AeST source gravitationally and MI does not. The completion problem is: *can the phantom be sourced without giving up the derivation of a_0 's scale?*

5 The completion problem

5.1 The desiderata

A satisfying MI completion of lensing would keep all of the following. We state each narrowly, so the theorem rests on definitions rather than on plausibility steps.

- **D (a_0 -derived).** a_0 enters the theory *solely* as the horizon-floored argument of the passive kernel $K(\Box_u/a_0^2)$ — **not** as an independent Lagrangian coupling. (D is the *scale-tie*; the value and sign remain postulates, Sec. 1.2.)
- **S (single-metric).** Photons and gravitons propagate on one metric, $c_\gamma = c_{\text{GW}}$.
- **G (ghost-free).** No Ostrogradsky mode, no negative-norm state.
- **L (MOND lensing).** The theory reproduces $g_{\text{lens}} = \nu(y)g_{\text{bar}} \rightarrow \sqrt{a_0 g_{\text{bar}}}$ ($F \rightarrow 1$): null geodesics feel the phantom $(\nu - 1)\rho$. *We state L without reference to the number of metrics* — this is what makes S a non-redundant, essential hypothesis (Sec. 7.2).

with two secondary constraints — Cassini (Solar-System PPN safety, $\nu - 1 \sim 1/(2y) \rightarrow 0$ at $y \sim 10^6$) and cosmology preservation (the growing mode $\nu_{\text{cosmo}} \in [1, 1.09]$, $\sigma_8 + 2\text{--}3\%$, from the horizon-floored argument).

5.2 Two lemmas

The theorem is carried by two universal lemmas; the candidate constructions of Sec. 6 are existence witnesses for the surviving corners, not the argument itself.

Lemma 1 (nonlocality — uses S). Under S , the lensing deflection is set by the single metric's stress tensor, so L requires the phantom $(\nu - 1)\rho$ to be **real gravitating stress-energy**. For a point mass M ($g_{\text{bar}} = GM/r^2$, $y = GM/(a_0 r^2)$) the enclosed phantom that makes $g_{\text{lens}} = \nu g_{\text{bar}}$ is $M_{\text{ph}}(r) = (\nu(y) - 1)M$, and (sympy-exact)

$$\frac{M_{\text{ph}}(r)}{r} \xrightarrow{r \rightarrow \infty} \sqrt{\frac{a_0 M}{G}} \neq 0, \quad (16)$$

an **unbounded halo** whose enclosed mass keeps growing where $\rho = 0$. The precise content: a *local* functional of ρ and the passive frame produces a phantom whose radial profile is fixed by the local acceleration invariant $|a|(r)$ **alone**, so its amplitude cannot carry the source mass M that $(\nu - 1)M$ requires — the same functional applied to two different masses gets the profile wrong for at least one (the mass-blindness that kills C1, Sec. 6.1). Matching the *mass-correct, universal* phantom for **all** sources therefore requires a nonlocal carrier — a term whose variation yields the elliptic equation $\nabla \cdot [\dots] = \rho$ (the inverse Laplacian of AQUAL/QUMOND). (A local term can produce a nonzero off-source phantom; what it cannot produce is the mass-correct one for every body.)

Lemma 2 (boundedness — uses passivity). By $\sup K = 1$, the passive kernel delivers only $O(K) \leq 1$ dressing. On the RAR shell it gives the uu coefficient $K = 1/\nu$ (suppression) and slip $2K'X/K = 1/(2y + 1)$ (bounded, tension). The phantom instead needs uu coefficient $\nu - 1/\nu = 1/\sqrt{y(y + 1)}$ (deep-MOND $\sim 1/\sqrt{y}$, unbounded); the shortfall

$$\frac{\text{needed}}{\text{supplied}} = \frac{\nu - 1/\nu}{1/\nu} = \nu^2 - 1 = \frac{1}{y} \quad (17)$$

diverges as $y \rightarrow 0$. No normalized passive kernel reaches ν on the RAR shell (a direct counterexample search finds the passive cap 1.0 against a deep-MOND target ≈ 31.6 ; the value the frame actually realizes, $1/\nu$, is the *farthest* point from ν).

Both footings: the halo amplitude $\sqrt{a_0 M/G}$ carries a_0 , but the nonlocality (the nonzero limit) and the shortfall $1/y$ are footing-independent — they hold for 9.36×10^{-11} and 1.13×10^{-10} identically.

6 Candidate mechanisms

We computed three explicit completions. Each is a genuine attempt to close the wedge; each is an existence witness landing on a different corner of the obstruction. All scripts exit 0 on both footings.

6.1 C1 — nonminimal frame-curvature: keeps D , fails L by mass-blindness

Add a frame-kernel-curvature coupling with no new dimensionful constant,

$$\Delta S = \frac{c^4}{16\pi G} \int \sqrt{-g} F(K(\Box_u/a_0^2)) Q, \quad Q \in \{R, u^\mu u^\nu R_{\mu\nu}\}. \quad (18)$$

Because a_0 enters only through X inside K and the prefactor is the bare $c^4/16\pi G$, this is a clean D -side witness. Scalar–Ricci variation gives the standard scalar-tensor form; the new gravitating source is $-(g_{\mu\nu}\Box - \nabla_\mu \nabla_\nu)F(K)$, whose static 00 projection is a Laplacian of a local scalar. By Gauss the enclosed phantom is a pure surface flux, $M_{\text{ph}}(r) = (c^2/G) \kappa r^2 F'(r)$ — nonzero off-source, so a local term *does* source a phantom. But imposing lensing closure $M_{\text{ph}} = (\nu - 1)M_{\text{bar}}$ on a point mass yields (sympy-exact)

$$F'(y) = -\frac{\nu(y) - 1}{2\kappa\sqrt{y}} \frac{\sqrt{GMa_0}}{c^2} \propto \sqrt{M}. \quad (19)$$

The required shape **carries the source mass**: $F'_{\text{req}}(y; M_2)/F'_{\text{req}}(y; M_1) = \sqrt{M_2/M_1}$, independent of y . A local $F(K)$ sees only $|a|$ — it is blind to the source mass — so no single universal F sources the phantom for more than one object (Lemma 1). Fixing F on a $6 \times 10^{10} M_\odot$ galaxy and applying it to a $10^{14} M_\odot$ cluster, the cluster receives phantom fraction $\sqrt{6 \times 10^{10}/10^{14}} \approx 0.024$, i.e. $\sim 2.4\%$ of the needed phantom — under-lensed by a factor ≈ 41 ; the reverse tuning over-lenses by the same factor. The miss is footing-independent (a_0 cancels in the ratio). Separately, under honest metric variation $X = |a|^2/a_0^2$ is a function of first metric derivatives, so $\Box F(K)$ contains third metric derivatives (Φ''') — a genuine Ostrogradsky ghost, unless one freezes X and prescribes the scalar by hand (no longer MI). Crucially, **C1 fails L by mass-blindness regardless of its ghost**, so it is a clean $D = 1$ witness for the “ G is not the lever” step. Verdict: **fails lensing, D kept**.

6.2 C2 — nonlocal-in-matter (Deffayet–Esposito–Farèse–Woodard class): closes L , loses D

Take a purely metric, single-metric, nonlocal-in-matter distortion of Einstein–Hilbert,

$$S = \frac{1}{16\pi G} \int \sqrt{-g} R [1 + f(\Box^{-1} R)] + S_{\text{matter}}[g, \psi], \quad (20)$$

with $\Box^{-1}$ retarded and f chosen so the weak-field limit is the QUMOND/AQUAL modified-Poisson equation. This is the nonlocal-metric MOND realization of Deffayet, Esposito-Farèse & Woodard [5] (background: [6]).

It works. The nonlocal term sources the phantom $\nu\rho$ on the single metric as a genuine scalar energy density with zero anisotropic stress; the no-slip theorem gives $\Phi = \Psi$, so $g_{\text{lens}} = \nu g_{\text{bar}}$ with the deep slope $\rightarrow \sqrt{a_0 g_{\text{bar}}}$ ($F \rightarrow 1$, machine-checked). It is purely metric, so $c_\gamma = c_{\text{GW}}$ (GW170817-safe [8]), and Cassini-safe by construction ($\nu - 1 = 5 \times 10^{-7}$ at $y = 10^6$).

But a_0 is now **free**, and the load-bearing reason is structural, not merely a fit degeneracy: there is no passive frame, no kernel $K(\Box_u)$, no $Z = \sqrt{32\pi/3}$ — the nonlocal operator acts on R

(matter curvature), not on a vacuum frame, so nothing ties a_0 to cH_Λ . It enters as an independent coupling inside f , i.e. $\neg D$ by definition. (Corroborating but weaker: the MOND relation is form-invariant under $(a_0, g_{\text{bar}}) \rightarrow \lambda(a_0, g_{\text{bar}})$, so both footings fit equally. This footing non-diagnostics is a property of the RAR relation itself and is shared by a_0 -derived pure MI — the SPARC RAR is likewise convention-compatible and non-diagnostic of 9.36×10^{-11} — so it does not by itself mark C2 as MG; the structural absence of the kernel does.) One residual partial: $\square^{-1}$ carries a horizon IR scale, so $a_0 \sim cH_\Lambda$ is parametrically *natural* — but natural is not derived, and Z is not sourced. Secondary costs: a localized would-be ghost (kinetic eigenvalues $\pm \frac{1}{2}$) tamed only by the contested retarded prescription, and a separate cosmology fit of f . Verdict: **closes L , a_0 free (=MG)**.

6.3 C3 — frame/scalar carrier: closes L , loses D , plus one new dof

Ask whether the frame’s dressed momentum can source curvature. Sub-candidate **C3a** (passive frame only, a_0 kept derived, 0 new dof): the matter leg’s uu coefficient is $O(K) = 1/\nu$ and the frame-constraint leg vanishes identically ($u \cdot u = -1$), so the assembled source delivers the suppression $1/\nu$, short of the needed ν by $\nu^2 - 1 = 1/y$, diverging deep-MOND. **Fails lensing**, reproducing the stress-tensor result of Sec. 4.2 exactly. Sub-candidate **C3b** (promote the carrier to a nonlocal scalar): the only local $h(X)$ reaching the deep phantom is $h \sim a_0/|a|$ — the AQUAL term — which makes the potential equation nonlinear and the phantom nonlocal (the same unbounded halo $M_{\text{ph}}/r \rightarrow \sqrt{a_0 M/G}$). A spatial-kinetic carrier is then mandatory, so it **propagates**: one new scalar dof, its acceleration scale a free coupling. Lensing closes ($F \rightarrow 1$), $c_\gamma = c_{\text{GW}}$ holds (the scalar sources via its own $T_{\mu\nu}$, no disformal photon coupling), ghost-freedom holds ($f' > 0$), Cassini-safe — but a_0 is **free** and cosmology is forfeited to the new scalar. Verdict: **closes L , a_0 free (=MG), +1 scalar dof**.

6.4 Scorecard

lane	L (lensing)	S ($c_\gamma = c_{\text{GW}}$)	G (ghost)	Cassini	a_0	verdict
C1	FAIL (mass-blind $\sqrt{M}$; 2.4% / 41 $\times$)	OK (R) / risk ($uuR_{\mu\nu}$)	FAIL (honest-var Ostrogradsky Φ''')	OK	kept-derived	fails lensing
C2	PASS ($F \rightarrow 1$, $\sqrt{}$ slope)	PASS (purely metric)	cost ($\pm \frac{1}{2}$; prescription-dependent)	PASS*	FREE	closes, a_0 free
C3a	FAIL (delivers $1/\nu$, short $1/y$)	PASS	PASS (0 dof)	PASS	kept-derived	fails lensing
C3b	PASS ($F \rightarrow 1$, $\sqrt{}$ slope)	PASS (own $T_{\mu\nu}$)	PASS ($f' > 0$)	PASS	FREE (+1 dof)	closes, a_0 free

*C2 inherits the AeST-class Solar-System Q_2 -quadrupole caveat [11, 12]. C1’s cosmology is plausible; C2 requires a separate fit; C3b forfeits cosmology to the new scalar.

The pattern is exact: every lane that keeps D (C1, C3a) fails L ; every lane that achieves L (C2, C3b) loses D . No lane holds both. C2’s G bit is prescription-dependent, but C2 lands on $\neg D$ independent of its G bit; C3b is the clean ghost-free L -side witness. These are witnesses; the rigor of the result lives in Lemmas 1–2, not in the enumeration.

7 The obstruction theorem

7.1 Statement

Theorem (conditional on kernel passivity). *Assume the passivity premise (Sec. 2). Then $D \wedge S \Rightarrow \neg L$: no field theory extending the MI action keeps a_0 as the passive-kernel argument (D), propagates on one metric (S), and reproduces the MOND-lensing phantom $(\nu - 1)\rho$ (L). Equivalently, $\{D, S, L\}$ is the **unique minimal unsatisfiable subset** of $\{D, S, G, L\}$: each proper subset is realized —*

- $\{D, S\} = \text{pure MI (the present theory; } L \text{ off)}$;
- $\{S, L\} = \text{Deffayet–Esposito–Farèse–Woodard / AeST with } a_0 \text{ a } \mathbf{free} \text{ coupling (} D \text{ off)}$;
- $\{D, L\} = \text{a disformal two-cone construction } \tilde{g} = g + B(\Box_u)u_\mu u_\nu \text{ with } B \text{ in the passive kernel, which keeps } a_0 \text{ derived but carries a second photon cone excluded observationally by GW170817 (} S \text{ off)}$.

Ghost-freedom G enters no step.

The connective is “cannot coexist,” $\neg(D \wedge L)$ under S — *not* exclusive-or: the pure-GR corner has neither D nor L , which an XOR would misclassify. Earlier notes that wrote “ $D \oplus L$ / XOR” are corrected here.

7.2 Why S is essential (and the theorem non-redundant)

Because L is stated metric-agnostically (Sec. 5.1), S is a genuine, independent hypothesis with a precise role: under S , lensing is fixed by the one metric’s $T_{\mu\nu}$, so L forces the phantom to be real gravitating stress-energy, and Lemma 1 then bites. Drop S — allow a disformal second cone $\tilde{g} = g + B uu$ — and photons see a *different* metric: the phantom becomes pure geometry with no new stress-energy, Lemma 1 is evaded, a_0 stays inside the passive kernel $B(\Box_u)$, and L is achieved with D kept. That $\{D, L, \neg S\}$ construction is logically satisfiable as a field theory; it is killed only *observationally*, by GW170817 [8]: a disformal coupling large enough to carry the $O(\nu)$ lensing deflection splits the photon and graviton cones far above the bound $|c_\gamma/c_{\text{GW}} - 1| \lesssim 10^{-15}$ — computed in full along the actual NGC 4993 sightline, the exclusion is ~ 6 – 7 orders of magnitude, robust across both footings [15, 9] — while the GW-safe small- B regime cannot source an $O(\nu)$ deflection: the same passivity/amplification wedge, now on the disformal axis. (A merely *conformal* bimetric $\tilde{g} = A(\phi)g$ preserves the null cone but cannot bend light beyond its baryonic content at all — the historical reason TeVeS [7] needed a disformal/vector part.) This is exactly why S belongs in the minimal unsatisfiable subset: the $\{D, L\}$ witness exists, and S is what excludes it.

7.3 The mechanism as lemmas, and the enumeration as bookkeeping

The theorem’s content is the single implication $D \wedge S \Rightarrow \neg L$, carried by:

- **Lemma 1 (nonlocality, Sec. 5.2, uses S):** a mass-correct universal single-metric L requires a nonlocal $(\nu - 1)\rho$ carrier.
- **Corollary (propagation $\Rightarrow \neg D$):** a nonlocal carrier reproducing the elliptic phantom has a spatial-kinetic term, so it propagates (a new dof) or its shape is a free function put in by hand; either way its acceleration scale is an independent Lagrangian coupling, i.e. **not** the passive-kernel argument — $\neg D$ by the definition of D (narrow- D makes this step a near-tautology rather than a plausibility claim). [Witnesses: C2, C3b.]

- **Lemma 2 (boundedness, Sec. 5.2, uses passivity):** keeping D confines the modification to the passive kernel $\sup K = 1$ ($O(K)$ dressing, $T_{00} = 0$ off $\text{supp } \rho$), which cannot source the $O(\nu)$ unbounded phantom — $\neg L$. [Witnesses: C1, C3a.]

A finite truth table over the 16 assignments of $\{D, S, G, L\}$ (script `nogo.py`, exit 0) then reads off that the target $D = S = L = 1$ is absent, that $\{D, S, L\}$ is the unique minimal unsatisfiable subset, and that the $\{D, L\}$ witness survives the mechanism but is GW-excluded. **The table is bookkeeping:** it is modus ponens over Lemmas 1–2, and the entire physics content lives in those two lemmas over the examined term classes (frame-curvature, nonlocal-in-matter, frame carrier). It is *not* an exhaustion over all possible Lagrangians, and the result’s strength is exactly the strength of Lemmas 1–2 as physics statements over those classes. A referee who exhibits a term class outside the lemmas’ reach would genuinely extend the boundary.

The observationally viable single-metric corners:

D	S	G	L	identity
1	1	1	0	pure MI / C3a — a_0 derived, ghost-free, under-lenses (the present theory)
1	1	0	0	C1 — local ghostly frame term; still fails L (mass-blind) even with a ghost
0	1	1	1	C2, C3b — closes L , single-metric, ghost-free, but a_0 free (=MG)
0	1	0	1	nonlocal with untamed ghost — closes L , a_0 free, ghost (contested)
0	1	1	0	GR / trivial
0	1	0	0	GR + ghost (vacuous)

The target $D = S = G = L = 1$ is absent, and so is its ghost-relaxed sibling $D = S = 1, G = 0, L = 1$: relaxing ghost-freedom does not rescue $\{D, S, L\}$, because C1 fails L by mass-blindness regardless of its ghost. This is why G is not the lever.

7.4 Footing independence

The wedge is a_0 -free. The shortfall $\nu^2 - 1 = 1/y$ and the mass-blindness ratio $\sqrt{M_2/M_1}$ are dimensionless in $y = g_{\text{bar}}/a_0$, and ν is form-invariant under $(a_0, g_{\text{bar}}) \rightarrow \lambda(a_0, g_{\text{bar}})$. Both footings ($9.36 \times 10^{-11} / 1.13 \times 10^{-10} \text{ m s}^{-2}$) give the identical dimensionless verdict.

8 The passivity/amplification dichotomy

Here is the physical crux, the intended central idea of the paper.

$a_0 = cH_\Lambda/Z$ is scale-derivable *only because* the MI modification is a passive, causal vacuum-response kernel with a normalized positive measure. Passivity floors the response amplitude at the horizon scale — it is the source of the scale-tie. But passivity has a mathematical consequence: $\sup K = 1$. A passive kernel can only **suppress**. On the RAR shell it delivers exactly ρ/ν ; deep-MOND $K \rightarrow 0$, the source dressed all the way down.

MOND lensing demands the opposite. The phantom $(\nu - 1)\rho$ is an **enhancement** to $\nu\rho$, with $\nu > 1$ — strictly outside the passive cone whose supremum is 1. The needed uu coefficient is $\nu - 1/\nu = 1/\sqrt{y(y+1)}$ (unbounded, deep $\sim 1/\sqrt{y}$); the passive kernel supplies $1/\nu$; the shortfall $\nu^2 - 1 = 1/y$ diverges as $y \rightarrow 0$. So the dichotomy is exact and structural, not a failed search:

The very property that ties a_0 ’s scale to the vacuum — passivity, $\sup K = 1$ — is the property that forbids the lensing phantom. Enhancement requires an anti-dissipative (pumped) kernel, whose amplitude is a free coupling, so a_0 would no longer be cH_Λ/Z .

The nonlocality lemma closes the loop from the other side: the mass-correct phantom of a point mass is an unbounded halo ($M_{\text{ph}}(r)/r \rightarrow \sqrt{a_0 M/G}$), so sourcing it universally forces a nonlocal carrier with a spatial-kinetic term — a propagating dof whose acceleration scale is a free coupling. Passivity forbids the enhancement; the halo’s nonlocality forces propagation; propagation frees a_0 . The two horns are the same wall seen from two sides.

Two honest guards on this statement. First, boundedness ($\neg L$) uses *only* $\sup K = 1$ — it does not invoke the a_0 value at all. So “ a_0 scale-tied” and “ $\neg L$ ” are not one claim chained back on itself; they are two consequences of the **one** passivity premise (a common-premise entailment, not a biconditional — we do not claim, and do not need, the converse “ a_0 -derived $\Rightarrow$ passive”). Second, this is sufficiency, not necessity: $\text{passivity} \Rightarrow (a_0 \text{ scale-tied} \wedge \neg L)$. Whether some non-passive kernel could also tie a_0 ’s scale — breaking the lock — is precisely the open pump door of Sec. 9, so the lock itself is conditional on staying inside the passive class.

9 Scope, conditionality, and consequences

We are explicit, both ways, about the strength of this result.

It is conditional. The obstruction rests on the single premise of MI-kernel passivity: the response is bounded, causal, positive-measure (Herglotz–Nevanlinna), $\|K\| \leq 1$, with the sum rule $\int d\mu/|t| = 1$, so its real-space dressing obeys $\sup K = 1$. This premise is banked and frozen [20, 21] but it is a *premise*, not a metatheorem. Were the kernel not passive, the argument would not run.

Within the passive class, D and $\neg L$ share one origin. The same passivity premise is what ties a_0 ’s scale and what caps K at 1; $\neg L$ follows from the cap alone (Sec. 8). One cannot, inside the passive class, keep the scale-tie while discarding the boundedness that forbids L — they are two faces of the one hypothesis, which is what makes the obstruction a real dichotomy rather than a coincidence of two separate assumptions.

The one MI escape, and why it is currently shut. The only way out that keeps a_0 tied to cH_Λ with a derived amplitude is a causal *enhancing* kernel ($\nu > 1$) — an anti-dissipative / pumped vacuum response. The banked sign walls [20, 21] find no such channel pump-free: passive, reactive, and 4-point/dS-positivity edges are all sign-blind or anti-MOND-locked, and the enhancement reopens only under a new forced pump mechanism that has not been exhibited. **We do not claim the theory is closed.** The door labeled “forced pump” remains open in principle; what the banked results establish is that it is not open by any pump-free route so far identified — and, since a non-passive a_0 -scale-tying kernel is exactly what would break the D /boundedness lock, that same door is the one place the lock could fail.

Consequences. (i) *A completion exists — as modified gravity, at the cost of the a_0 scale-tie.* C2 and C3b constructively demonstrate a single-metric, GW170817-safe field theory reproducing MOND dynamics and MOND lensing with the correct deep slope — real, honest content — but it is modified gravity in the Deffayet–Esposito–Farèse–Woodard / AeST class [5, 4], forfeiting the vacuum-tied $a_0 = cH_\Lambda/Z$, plus (C2) a separate cosmology fit and a contested would-be ghost, or (C3b) one new propagating scalar and a forfeited cosmology. The cost is precisely the program’s single most distinctive claim. (ii) *Lensing becomes a shared-MG observable.* Because the only completions that reproduce the lensing phantom are MG, weak lensing can no longer discriminate MI from MG — both predict the same $F \rightarrow 1$ relation. The clean modified-gravity-impossible discriminant that remains is the non-adiabatic relational velocity-dispersion spread ($\sim 6\text{--}13\%$ in MI from frame-history-dependent inertia; exactly zero in MG) — currently underpowered; it is developed separately in Ref. [26] and only pointed to here. Any MG completion also inherits the AeST-class Solar-System Q_2 -quadrupole caveat [11, 12], a shared-MG cost, honestly stated. (iii)

The obstruction shows the a_0 reframing is load-bearing. If $a_0 = cH_\Lambda/Z$ were a cosmetic relabeling, one could keep it while adding any lensing structure. That it is incompatible — within the passive class — with single-metric MOND lensing shows it is a genuine, restrictive, physical commitment. Far from a funeral, this is evidence the reframing is doing work.

10 Conclusion

We have presented a single covariant action for the de Sitter–Unruh modified-inertia program — Einstein–Hilbert gravity, a passive unit-timelike frame with a machine-verified second-class Dirac closure keeping it at zero propagating degrees of freedom, and a matter sector dressed by a rigorously defined Herglotz–Nevanlinna kernel — ghost-free and well posed conditional on the passive-frame premise, with the ring-exact reduction to Milgrom’s interpolating law as a derivation from the first-moment identity $u \cdot \square_u u = -|a|^2$ rather than an ansatz. Within this theory we have identified, and machine-verified on both footings, an obstruction — conditional on kernel passivity: **a passive-kernel-tied a_0 and single-metric MOND weak lensing cannot coexist.** The central implication is $D \wedge S \Rightarrow \neg L$; equivalently $\{D, S, L\}$ is the unique minimal unsatisfiable subset of $\{D, S, G, L\}$, with $\{D, S\} = \text{pure MI}$, $\{S, L\} = \text{Deffayet–Esposito–Farèse–Woodard / AeST (a_0 free)}$, and $\{D, L\} = \text{a disformal two-cone construction (a_0 kept) killed observationally by GW170817}$. Ghost-freedom is not the lever. The mechanism is a passivity/amplification dichotomy reinforced by a nonlocality lemma; three explicit candidates witness the surviving corners. The result is conditional on kernel passivity — but that premise is the very thing that ties a_0 ’s scale, so within the passive class the scale-tie and the obstruction are two faces of one hypothesis. We do not claim the theory is closed. The honest map: the MI program is complete for dynamics (0.108 dex on SPARC) and viable in cosmology, and its lensing sector completes only by crossing into modified gravity at the cost of the a_0 reframing. Lensing is now a shared observable; the clean discriminant is the relational σ -spread [26]. And the obstruction itself is the strongest evidence that the $a_0 = cH_\Lambda/Z$ reframing is load-bearing physics rather than a relabeling.

Data and code availability

The repository <https://github.com/carlzimmerman/zimmerman-formula> is self-verifying: every load-bearing numerical claim in this paper traces to a committed, runnable Python script (sympy/numpy) that exits 0 on both a_0 footings, with adversarial VERIFY documents in the source lanes and no hard-coded verdict booleans (grep-audited). The obstruction-theorem scripts live in `prep_2026/mi_lensing_completion/` (Appendix A); the field-theory scripts in `prep_2026/mi_field_theory/` and `mi_closure_pin/`. The companion tool `a0kit` reproduces the phenomenology and is archived at Zenodo, DOI [10.5281/zenodo.21478981](https://doi.org/10.5281/zenodo.21478981) (ASCL submission pending editor review). The consolidated source deposits are Refs. [14, 15, 16, 17, 18]. SPARC data are read from the repository’s frozen copies of the public Lelli et al. release [13]; KiDS-1000 lensing from Brouwer et al.’s machine-readable release [10].

A Verification scripts

All scripts are committed in `prep_2026/mi_lensing_completion/` (theorem) and `prep_2026/mi_field_theory/`, `mi_closure_pin/` (construction), run against a frozen read-only repository, and exit 0 on both footings. They use sympy for exact symbolic checks and numeric grids where a radical will not auto-collapse. No candidate passes all checks, so none is asserted as a full completion.

- `rederive_identity.py`, `closure_map.py` — the first-moment identity $u \cdot \square_u u = -|a|^2$ (three routes) and the ring-exact collapse to $\nu = \sqrt{1 + 1/y}$ (residual $\sim 10^{-13}$).
- `matter_coupling_Tmunu.py` (14/14) — the stress tensor $T = \alpha uu + \beta g + \gamma aa$, WEP $\eta < 10^{-12}$, conservation residual 2.6×10^{-16} .
- `wellposed.py` (28/28) — dof census, Dirac second-class closure (determinant $\rightarrow 4$), hyperbolicity, causality.
- `mi_closure_pin/ostro_nonlocal_verify.py` (13/13) — the genuine auxiliary-tower Herglotz ghost check with live negative controls.
- `c1_frame_curvature.py` (11/11) — $M_{\text{ph}}(r) = (c^2/G)\kappa r^2 F'(r)$; lensing closure forces $F' \propto \sqrt{M}$ (sympy-exact); the $\sqrt{M_2/M_1}$ mass-blindness; the cluster fraction ≈ 0.024 ($\sim 41\times$); the honest-variation Ostrogradsky term.
- `c2_nonlocal.py` (18/0) — the DEW term sources $\nu\rho$ with zero anisotropic stress, no-slip, $F \rightarrow 1$ with the $\sqrt{a_0 g_{\text{bar}}}$ slope; $c_\gamma = c_{\text{GW}}$; the $\pm \frac{1}{2}$ would-be-ghost eigenvalues; form-invariance under $(a_0, g_{\text{bar}}) \rightarrow \lambda(\dots)$.
- `c3_carrier.py` — C3a delivers the $1/\nu$ suppression (shortfall $1/y$); C3b's local $h \sim a_0/|a|$ gives the unbounded halo, forcing a propagating scalar with a_0 free.
- `nogo.py` — the completeness check: the nonlocality limit; the boundedness shortfall and the tension-signed slip leg; the mass-blindness ratio; footing independence; and the finite truth table over $\{D, S, G, L\}$ showing $D \wedge S \Rightarrow \neg L$ and that $\{D, S, L\}$ is the unique minimal unsatisfiable subset.
- `verify_adversarial.py` — an independent re-derivation (not reusing the candidate assertions) of the wedge ν^2 , the dichotomy, the mass-blindness $\sqrt{M}$, a direct passive-kernel counterexample search, and a check that C2's $F \rightarrow 1$ is a genuine (not manufactured) closure that is nonetheless MG with a_0 free. Trap sweep both directions: no manufactured completion, no manufactured no-go. Verdict: upheld.

B The derived/postulated ledger (condensed)

status	item
Derived	the Herglotz–Nevanlinna measure of K (unique, closed-form densities, sum rule = 1); the first-moment identity and the ring-exact reduction to $\nu = \sqrt{1 + 1/y}$; the stress tensor and its conservation; WEP $\eta = 0$; the 0-dof Dirac closure of the frame; $c_T = 1$; the two lemmas and the obstruction theorem (conditional on passivity); footing independence of the wedge.
Postulated	the value of a_0 ($\kappa = 1/2$ provably unforceable); $Z = \sqrt{32\pi/3}$; the sign $s = -1$; the passive/hypersurface-orthogonal frame premise (the load-bearing hinge); kernel passivity itself (the theorem’s single premise).
Asserted, not verified	the fully coupled all-orders nonlocal Hamiltonian; global $B < 1$ off spherical symmetry for the disformal witness. (The auxiliary-tower Herglotz ghost check <i>is</i> machine-verified, 13/13 with live negative controls.)
Open	the “forced pump” door (a non-passive kernel that ties a_0 ’s scale); term classes outside the two lemmas’ reach.

References

- [1] M. Milgrom, “A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis,” *Astrophys. J.* **270**, 365 (1983).
- [2] M. Milgrom, “The modified dynamics as a vacuum effect,” *Phys. Lett. A* **253**, 273 (1999), arXiv:astro-ph/9805346. [The identical ν -kernel modified-inertia law, Eq. (9); his coefficient was $2cH_\Lambda$.]
- [3] C. M. Limbach, D. Psaltis, and F. Özel, “The redshift evolution of the Tully–Fisher relation as a test of modified gravity,” arXiv:0809.2790 (2008). [The prior $a_0 \propto \sqrt{\rho_{\text{DE}}}$ scale coincidence.]
- [4] C. Skordis and T. Zlosnik, “New relativistic theory for modified Newtonian dynamics,” *Phys. Rev. Lett.* **127**, 161302 (2021), arXiv:2007.00082. [AeST; the single-metric MG comparison class.]
- [5] C. Deffayet, G. Esposito-Farèse, and R. P. Woodard, “Nonlocal metric formulations of MOND with sufficient lensing,” *Phys. Rev. D* **84**, 124054 (2011), arXiv:1106.4984.
- [6] C. Deffayet and R. P. Woodard, “Reconstructing the distortion function for nonlocal cosmology,” *JCAP* **08**, 023 (2009), arXiv:0904.0961.
- [7] J. D. Bekenstein, “Relativistic gravitation theory for the modified Newtonian dynamics paradigm,” *Phys. Rev. D* **70**, 083509 (2004).
- [8] B. P. Abbott et al. (LIGO Scientific, Virgo, Fermi-GBM, INTEGRAL), “Gravitational waves and gamma-rays from a binary neutron star merger: GW170817 and GRB 170817A,” *Astrophys. J. Lett.* **848**, L13 (2017), arXiv:1710.05834.
- [9] S. Boran, S. Desai, E. O. Kahya, and R. P. Woodard, “GW170817 falsifies dark matter emulators,” *Phys. Rev. D* **97**, 041501 (2018), arXiv:1710.06168.

- [10] M. M. Brouwer et al., “The weak lensing radial acceleration relation: constraining modified gravity and cold dark matter theories with KiDS-1000,” *Astron. Astrophys.* **650**, A113 (2021), arXiv:2106.11677.
- [11] H. Desmond, A. Hees, and B. Famaey, “On the tension between the radial acceleration relation and Solar System quadrupole in modified gravity MOND,” *Mon. Not. R. Astron. Soc.* **530**, 1781 (2024).
- [12] M. Park et al., “Solar-System quadrupole constraints on modified gravity,” arXiv:2602.17884 (2026).
- [13] F. Lelli, S. S. McGaugh, and J. M. Schombert, “SPARC: Mass models for 175 disk galaxies with Spitzer photometry and accurate rotation curves,” *Astron. J.* **152**, 157 (2016), arXiv:1606.09251.
- [14] C. P. Zimmerman, “A de Sitter–Unruh modified-inertia field theory, complete up to its constants,” Zenodo (2026), DOI [10.5281/zenodo.21403470](https://doi.org/10.5281/zenodo.21403470).
- [15] C. P. Zimmerman, erratum to Ref. [14] (the GW170817 disformal-sector exclusion, computed in full), Zenodo (2026), DOI [10.5281/zenodo.21415677](https://doi.org/10.5281/zenodo.21415677).
- [16] C. P. Zimmerman, “The covariant modified-inertia completion” (concept record, versions v1–v11), Zenodo (2026), concept DOI [10.5281/zenodo.21253644](https://doi.org/10.5281/zenodo.21253644).
- [17] C. P. Zimmerman, “A passivity obstruction: why a derived $a_0 = cH_\Lambda/Z$ and single-metric weak lensing cannot coexist in de Sitter–Unruh modified inertia,” Zenodo (2026), DOI [10.5281/zenodo.21418816](https://doi.org/10.5281/zenodo.21418816).
- [18] C. P. Zimmerman, “The de Sitter–Unruh a_0 no-go” (adversarially reviewed honest-scope paper), Zenodo (2026), DOI [10.5281/zenodo.20973740](https://doi.org/10.5281/zenodo.20973740). [Its item (iii) covariant-completion no-go is superseded here; see Sec. 1.3.]
- [19] C. P. Zimmerman, “The MI action, v3” (correction: the Einstein-aether strong-coupling wall does not apply to the passive 0-dof frame), Zenodo (2026), DOI [10.5281/zenodo.21263846](https://doi.org/10.5281/zenodo.21263846).
- [20] C. P. Zimmerman, “The MI action, v4” (Dirac closure; Herglotz kernel; nonlinear stability), Zenodo (2026), DOI [10.5281/zenodo.21264727](https://doi.org/10.5281/zenodo.21264727).
- [21] C. P. Zimmerman, “The MI action, v11” (the one-loop dS edge at divergence level; a_0 unrenormalized; sum rule), Zenodo (2026), DOI [10.5281/zenodo.21284144](https://doi.org/10.5281/zenodo.21284144).
- [22] C. P. Zimmerman, “The parameter-free $a_0(z)$ test,” Zenodo (2026), DOI [10.5281/zenodo.20737162](https://doi.org/10.5281/zenodo.20737162). [JCAP-side companion.]
- [23] C. P. Zimmerman, “The non-circular cross-scale $a_0(z)$ test,” Zenodo (2026), concept DOI [10.5281/zenodo.21440407](https://doi.org/10.5281/zenodo.21440407). [JCAP-side companion.]
- [24] C. P. Zimmerman, “Rubin/LSST pre-registration of the $a_0(z)$ test,” Zenodo (2026), DOI [10.5281/zenodo.21478568](https://doi.org/10.5281/zenodo.21478568). [JCAP-side companion.]
- [25] C. P. Zimmerman, “The a_0 -line: a gas-dominated SPARC slope measurement and the Λ inversion,” Zenodo (2026), DOI [10.5281/zenodo.21419735](https://doi.org/10.5281/zenodo.21419735). [MNRAS-side companion.]

- [26] C. P. Zimmerman, “The relational velocity-dispersion spread as an MI-distinctive observable,” Zenodo (2026), DOI [10.5281/zenodo.21421896](https://doi.org/10.5281/zenodo.21421896).
- [27] C. P. Zimmerman, `a0kit`: the a_0 -line phenomenology toolkit (software), Zenodo (2026), concept DOI [10.5281/zenodo.21478981](https://doi.org/10.5281/zenodo.21478981). ASCL submission pending editor review.